Plotting Magnetic Fields

Exam code: 0625 & 0972

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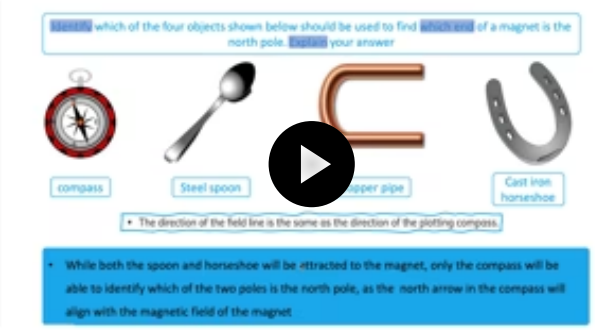

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Plotting magnetic field lines

- In your IGCSE examination, you might be asked to describe a method of plotting the magnetic field around a bar magnet
- There are two principal ways of doing this:
  - using iron filings
  - using plotting compasses

Using iron filings

- Place a piece of paper on top of the magnet
- Gently sprinkle iron filings on top of the paper
- Now carefully tap the paper to allow the iron filings to settle on the field lines

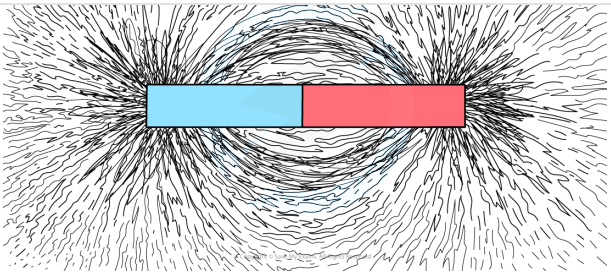
Magnetic field lines using iron filings

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

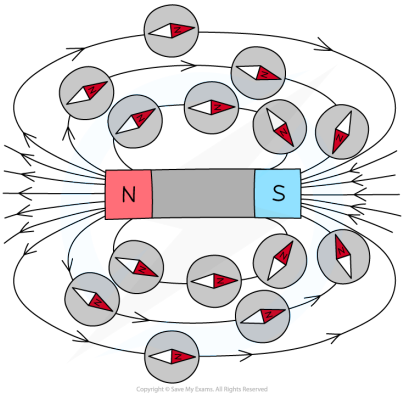


Iron filings can be used to plot a magnetic field

Using plotting compasses

- Place the magnet on top of a piece of paper
- Draw a dot at one end of the magnet (near its corner)
- Place a plotting compass next to the dot, so that one end of the needle of the compass points towards the dot
- Use a pencil to draw a new dot on the other side of the compass needle
- Now move the compass so that it points towards the new dot, and repeat the above process
- Keep repeating until you have a chain of dots going from one end of the magnet to the other. Then remove the compass, and link the dots using a smooth curve. This is the magnetic field line
- The direction of the field lines go from the **north pole** to the **south pole** of the magnet.
    - This is the same as the direction of the north arrow on the plotting compass, which points towards the south pole
- Repeat the process to create other magnetic field lines around the bar magnet

Using a compass to create magnetic field lines



Compasses can be used to plot the magnetic field around a bar magnet

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